Supplementary Material: Exact Structural Abstraction and Tractability Limits

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
FR1	currentFamilies_partition_by_role Paper4dFrontier/Classification.lean	FR2	currentFamilies_partition_counts Paper4dFrontier/Classification.lean
FR3	current_families_have_finite_basis Paper4dFrontier/Classification.lean	FR4	current_positive_interfaces_factor_through_role Paper4dFrontier/Classification.lean
FR5	lifted_families_reduce_to_core Paper4dFrontier/Classification.lean	FR6	degenerateFamilies_complete Paper4dFrontier/Classification.lean
FR7	exists_decisionProblem_realizing_labeling Paper4dFrontier/Realizability.lean	FR8	realizingProblem_decisionEquiv_iff Paper4dFrontier/Realizability.lean
FR13	isOptimal_positiveAffineTransform_iff Paper4dFrontier/FamilyAxioms.lean	FR14	opt_eq_positiveAffineTransform Paper4dFrontier/FamilyAxioms.lean
FR15	isSufficient_positiveAffineTransform_iff Paper4dFrontier/FamilyAxioms.lean	FR16	isRelevant_positiveAffineTransform_iff Paper4dFrontier/FamilyAxioms.lean
FR17	decisionEquiv_relabelActions_iff Paper4dFrontier/FamilyAxioms.lean	FR18	decisionEquiv_relabelStates_iff Paper4dFrontier/FamilyAxioms.lean
FR19	isSufficient_relabelActions_iff Paper4dFrontier/FamilyAxioms.lean	FR20	isSufficient_relabelStates_iff Paper4dFrontier/FamilyAxioms.lean
FR21	allProblems_relabelInvariant Paper4dFrontier/FamilyAxioms.lean	FR22	allProblems_kernelUniversal Paper4dFrontier/FamilyAxioms.lean
FR23	relabelInvariant_not_enough Paper4dFrontier/FamilyAxioms.lean	FR24	optRangeCard_realizingIdentity Paper4dFrontier/FamilyAxioms.lean
FR25	currentFamilies_explicit Paper4dFrontier/Classification.lean	FR26	coreFamilies_explicit Paper4dFrontier/Classification.lean
FR27	liftedFamilies_explicit Paper4dFrontier/Classification.lean	FR28	degenerateFamilies_explicit Paper4dFrontier/Classification.lean
FR29	primitiveMechanisms_explicit Paper4dFrontier/Classification.lean	FR30	isSufficient_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR31	isRelevant_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR32	isIrrelevant_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR33	isMinimalSufficient_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR34	sufficientSets_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR35	relevantSet_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR37	quotientCard_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR38	quotientCard_realizingIdentity Paper4dFrontier/FamilyAxioms.lean	FR39	realizingProblemQuotientEquivLabelRange_apply_quotientMap Paper4dFrontier/Realizability.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR40	realizingProblem_quotientCard_eq_labelRangeCard Paper4dFrontier/Realizability.lean	FR41	certificationStatistic_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR42	sufficientSetCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR43	minimalSufficientSetCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR44	relevantCoordCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR45	setoidRealizingProblem_decisionEquiv_iff Paper4dFrontier/Realizability.lean
FR46	setoidRealizingProblemQuotientEquivSetoidQuotient_apply_quotientMap Paper4dFrontier/Realizability.lean	FR47	exists_decisionProblem_realizing_setoid Paper4dFrontier/Realizability.lean
FR48	isSufficient_iff_agreementRel_le_decisionEquiv Paper4dFrontier/FamilyAxioms.lean	FR49	isIrrelevant_iff_sufficient_erase Paper4dFrontier/FamilyAxioms.lean
FR50	isRelevant_iff_not_sufficient_erase Paper4dFrontier/FamilyAxioms.lean	FR51	quotientEquiv_of_decisionEquiv_iff_apply_quotientMap Paper4dFrontier/FamilyAxioms.lean
FR52	empty_sufficient_of_constant_opt Paper4dFrontier/ClosureLaws.lean	FR53	irrelevant_of_constant_opt Paper4dFrontier/ClosureLaws.lean
FR54	decisionEquiv_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR55	isSufficient_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean
FR56	isRelevant_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR57	decisionEquiv_duplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR58	isSufficient_duplicateState_iff Paper4dFrontier/ClosureLaws.lean	FR59	isRelevant_duplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR60	duplicateStateQuotientEquivOriginal_apply_quotientMap Paper4dFrontier/ClosureLaws.lean	FR66	booleanCube_card Paper4dFrontier/EnumerationBounds.lean
FR67	exists_booleanCube_larger_than_monomial Paper4dFrontier/EnumerationBounds.lean	FR68	decisionEquiv_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR69	isSufficient_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean	FR70	isRelevant_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR71	addDuplicateStateQuotientEquivOriginal_apply_quotientMap Paper4dFrontier/ClosureLaws.lean	FR72	some_mem_opt_addDuplicateAction_iff Paper4dFrontier/ClosureLaws.lean
FR73	none_mem_opt_addDuplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR83	isSufficient_noiseExtended_iff Paper4dFrontier/DimensionalNoiseExtension.lean
FR84	isRelevant_noiseExtended_iff Paper4dFrontier/DimensionalNoiseExtension.lean	FR85	lastCoord_irrelevant_noiseExtended Paper4dFrontier/DimensionalNoiseExtension.lean
FR100	parentTree_realTreewidth_one Paper4dFrontier/ParentTreewidth.lean	FR101	low_rank_only_witness Paper4dFrontier/LowRankOnlyWitness.lean
FR105	separable_only_witness Paper4dFrontier/SeparableOnlyWitness.lean	FR106	bounded_actions_only_witness Paper4dFrontier/BoundedActionsOnlyWitness2.lean
FR108	all_independent_core_mechanisms_nondegenerate Paper4dFrontier/Block2Progress.lean	FR110	parentTreeStructured_implies_treeStructured Paper4dFrontier/ParentTreewidth.lean
FR111	treeStructured_and_unique_parent_iff_parentTreeStructured Paper4dFrontier/ParentTreewidth.lean	FR112	bounded_treewidth_only_witness Paper4dFrontier/BoundedTreewidthOnlyWitness.lean
FR113	symmetry_only_witness Paper4dFrontier/SymmetryOnlyWitness.lean	FR114	weak_tree_structured_not_width_one Paper4dFrontier/LegacyTreeGap.lean
FR115	tree_structure_hardness_bridge Paper4dFrontier/TreeStructureHardnessBridge.lean	FR118	pairCrossDifference_eq_binaryCrossDifference_of_lt Paper4dFrontier/BinaryPairwiseDichotomy.lean
FR119	pairwise_zero_crossDifference_unaryDecomposition Paper4dFrontier/BinaryPairwiseDichotomy.lean	FR120	binary_pairwise_symmetry_dichotomy Paper4dFrontier/BinaryPairwiseDichotomy.lean
FR121	actionGapCrossDifference_eq_binaryCrossDifference_of_lt Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR122	decisionRelevant_zero_actionGap_implies_unaryReduction Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR123	binary_pairwise_symmetry_decision_relevant_dichotomy Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR124	block6_genuine_interaction_dichotomy_obstruction Paper4dFrontier/Block6Obstruction.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR125	decisionRelevantInteractionGraph_addActionOffset Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR126	action_offset_can_force_constant_optimizer Paper4dFrontier/Block6Obstruction.lean
FR127	block6_action_offset_obstruction Paper4dFrontier/Block6Obstruction.lean	FR128	offsetNormalizedDecisionRelevantInteractionGraph_wellDefined Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR129	block6_offset_normalized_obstruction Paper4dFrontier/Block6Obstruction.lean	FR130	neverOptimalGhost_decisionRelevantGraph_eq_top Paper4dFrontier/Block6Obstruction.lean
FR131	neverOptimalGhost_supportedGraph_eq_bot Paper4dFrontier/Block6Obstruction.lean	FR132	support_filtering_removes_known_block6_obstructions Paper4dFrontier/Block6Obstruction.lean
FR133	block6_ghost_action_obstruction Paper4dFrontier/Block6Obstruction.lean	FR134	marginMasking_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean
FR135	marginMasking_zero_sufficient Paper4dFrontier/Block6Obstruction.lean	FR136	block6_optimizer_supported_obstruction Paper4dFrontier/Block6Obstruction.lean
FR137	MarginBounded Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR138	dominantPair_marginBounded Paper4dFrontier/Block6Obstruction.lean
FR139	dominantPair_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean	FR140	block6_margin_bounded_obstruction Paper4dFrontier/Block6Obstruction.lean
FR145	not_collapseLandscapeInfinity_of_oracle_predicate Paper4dFrontier/MetaCharacterization.lean	FR176	DecisionQuotient.DecisionProblem.quotient_is_coarsest paper4/DecisionQuotient/Quotient.lean
FR177	DecisionQuotient.DecisionProblem.quotient_has_unique_factorization paper4/DecisionQuotient/Quotient.lean	FR178	DecisionQuotient.DecisionProblem.srank_eq_relevant_card paper4/DecisionQuotient/Tractability/StructuralRank.lean
FR179	DecisionQuotient.quotientEntropy_le_srank_binary paper4/DecisionQuotient/Information.lean	FR180	DecisionQuotient.Statistics.fisherMatrix_rank_eq_srank paper4/DecisionQuotient/Statistics/FisherInformation.lean
FR181	DecisionQuotient.Information.compression_below_srank_fails paper4/DecisionQuotient/Information/RDSrank.lean	FR182	DecisionQuotient.Information.srank_bits_sufficient paper4/DecisionQuotient/Information/RDSrank.lean
FR183	DecisionQuotient.Physics.single_future_zero_cost paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean	FR184	DecisionQuotient.Physics.transportCost_pos_of_offDiag paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean
FR185	no_closureInvariant_predicate_of_orbit_gap Paper4dFrontier/ObstructionPredicateCandidates.lean	FR186	no_admissibleNormalizationPredicate_decides_dominantPair Paper4dFrontier/ObstructionPredicateCandidates.lean
FR187	no_admissibleNormalizationPredicate_decides_marginBounded Paper4dFrontier/ObstructionPredicateCandidates.lean	FR188	no_admissibleNormalizationPredicate_decides_ghostActionTwoPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean
FR189	no_admissibleNormalizationPredicate_decides_offsetActionZeroPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean	FR190	admissibleCollapseLandscapeInfinity_full_paper Paper4dFrontier/ObstructionPredicateCandidates.lean
FR191	ClosureSoundPackage Paper4dFrontier/ObstructionPredicateCandidates.lean	FR192	ReasonableGuardrailPackage Paper4dFrontier/ObstructionPredicateCandidates.lean
FR193	no_closureSoundPackage_predicate_decides_four_obstruction_families Paper4dFrontier/ObstructionPredicateCandidates.lean	FR194	no_reasonableGuardrailPackage_predicate_decides_four_obstruction_families Paper4dFrontier/ObstructionPredicateCandidates.lean
FR197	classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/AdmissibleCharacterization.lean	FR198	no_correctOnDomain_classifier_of_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean
FR199	correct_classifier_inherits_closureLawInvariant Paper4dFrontier/AdmissibleCharacterization.lean	FR200	admissibleNormalizationPredicate_has_explicit_inhabitants Paper4dFrontier/AdmissibleCharacterization.lean
FR201	closureLawInvariant_of_iff_of_closureEquivalent Paper4dFrontier/AdmissibleCharacterization.lean	FR202	exists_orbit_gap_of_not_closureLawInvariant Paper4dFrontier/AdmissibleCharacterization.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR203	closureLawInvariant_iff_no_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean	FR204	no_exact_closureLawInvariant_classifier_iff_ exists_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean
FR205	distinctActionCount Paper4dFrontier/DistinctActionProfiles.lean	FR206	actionCount_profileCompressedSlice Paper4dFrontier/DistinctActionProfiles.lean
FR207	decisionEquiv_profileCompressedSlice_iff Paper4dFrontier/DistinctActionProfiles.lean	FR208	profileCompressedSlice_preserves_ exactCertification Paper4dFrontier/DistinctActionProfiles.lean
FR209	profileCompressedSlice_bounded_actions Paper4dFrontier/DistinctActionProfiles.lean	FR210	OrbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean
FR211	no_orbitGapOn_of_exact_classifiable_by_ closureLawInvariant_onDomain Paper4dFrontier/AdmissibleCharacterization.lean	FR212	exact_classifiable_by_closureLawInvariant_ onDomain_iff_no_orbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean
FR213	no_exact_closureLawInvariant_classifier_ onDomain_iff_orbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean	FR214	boundedPatternScheme_holds_largeActionCount_ iff Paper4dFrontier/AdmissibleCharacterization.lean
FR215	boundedPatternDefinable_eventually_constant_ in_actionCount Paper4dFrontier/AdmissibleCharacterization.lean	FR216	boundedPatternDefinable_largeActionCount_ agrees Paper4dFrontier/AdmissibleCharacterization.lean
FR217	payloadSufficient_iff_realizingProblem_ isSufficient Paper4dFrontier/Realizability.lean	FR218	payloadRelevant_iff_realizingProblem_ isRelevant Paper4dFrontier/Realizability.lean
FR219	payloadIrrelevant_iff_realizingProblem_ isIrrelevant Paper4dFrontier/Realizability.lean	FR220	payloadMinimalSufficient_iff_realizingProblem_ isMinimalSufficient Paper4dFrontier/Realizability.lean
FR221	payloadSufficientSets_eq_realizingProblem_ sufficientSets Paper4dFrontier/Realizability.lean	FR222	payloadRelevantSet_eq_realizingProblem_ relevantSet Paper4dFrontier/Realizability.lean
FR223	feasiblePayloadSufficient_iff_ lawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR224	feasiblePayloadRelevant_iff_ lawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR225	feasiblePayloadIrrelevant_iff_ lawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR226	setValuedPayloadSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR227	setValuedPayloadRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR228	setValuedPayloadIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR229	outputSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR230	outputSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR231	outputSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR232	approximationSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR233	approximationSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR234	approximationSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR235	outputSemanticsSufficientSets_eq_of_ outputSemanticsEquivalent Paper4dFrontier/Realizability.lean	FR236	outputSemanticsRelevantSet_eq_of_ outputSemanticsEquivalent Paper4dFrontier/Realizability.lean
FR237	quotientMap_realizes_setoid_asOutputSemantics Paper4dFrontier/Realizability.lean	FR240	exists_outputSemantics_realizing_setoid Paper4dFrontier/Realizability.lean
FR241	outputSemanticsSetoid Paper4dFrontier/Realizability.lean	FR242	SemanticallyExtensionalMap Paper4dFrontier/Realizability.lean
FR243	semanticallyExtensionalMap_factors_through_ outputSemanticsQuotient Paper4dFrontier/Realizability.lean	FR244	semanticallyExtensionalClaim_factors_through_ outputSemanticsQuotient Paper4dFrontier/Realizability.lean
FR245	optimizerComputation_polytime_ closureLawInvariant Paper4dFrontier/ComputeCostApplications.lean	FR246	optimizerSetPayload_polytime_ closureLawInvariant Paper4dFrontier/ComputeCostApplications.lean
FR247	optimizerSetSearch_polytime_ closureLawInvariant Paper4dFrontier/ComputeCostApplications.lean	FR248	optimizerComputation_polytime_classifier_ agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostApplications.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR249	optimizerComputation_polytime_classifier_ agrees_on_closureEquivalent Paper4dFrontier/ComputeCostApplications.lean	FR250	no_correct_optimizerComputation_polytime_ classifier_decides_dominantPair Paper4dFrontier/ComputeCostApplications.lean
FR251	no_admissibleNormalizationPredicate_ optimizerComputation_polytime_and_dominantPair Paper4dFrontier/ComputeCostApplications.lean	FR252	CorrectnessForcesOrbitAgreementOnDomain Paper4dFrontier/AdmissibleCharacterization.lean
FR253	no_orbitGapOn_of_correct_classifier_onDomain_ of_forcedOrbitAgreement Paper4dFrontier/AdmissibleCharacterization.lean	FR254	correct_classifier_onDomain_iff_no_orbitGapOn_ of_forcedOrbitAgreement Paper4dFrontier/AdmissibleCharacterization.lean
FR255	no_correct_classifier_onDomain_iff_orbitGapOn_ of_forcedOrbitAgreement Paper4dFrontier/AdmissibleCharacterization.lean	FR256	optimizerComputation_polytime_correct_ classifier_onDomain_iff_no_orbitGapOn Paper4dFrontier/ComputeCostApplications.lean
FR259	optimizerSetPayload_polytime_classifier_ agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostApplications.lean	FR260	optimizerSetPayload_polytime_classifier_ agrees_on_closureEquivalent Paper4dFrontier/ComputeCostApplications.lean
FR261	optimizerSetSearch_polytime_classifier_agrees_ on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostApplications.lean	FR262	optimizerSetSearch_polytime_classifier_agrees_ on_closureEquivalent Paper4dFrontier/ComputeCostApplications.lean
FR263	relevant_of_uniformApprox_of_strict_gap_ witness Paper4dFrontier/ApproximateAdmissibility.lean	FR264	relevance_can_flip_under_arbitrarily_small_ uniform_perturbation Paper4dFrontier/ApproximateAdmissibility.lean
FR266	not_sufficient_of_uniformApprox_of_strict_gap_ witness Paper4dFrontier/ApproximateAdmissibility.lean	FR267	sufficiency_can_flip_under_arbitrarily_small_ uniform_perturbation Paper4dFrontier/ApproximateAdmissibility.lean
FR268	pacGuaranteeSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR269	pacGuaranteeSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR270	pacGuaranteeSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR271	regretGuaranteeSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR272	regretGuaranteeSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR273	regretGuaranteeSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR274	statisticalRiskSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR275	statisticalRiskSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR276	statisticalRiskSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR277	anytimeGuaranteeSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR278	anytimeGuaranteeSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR279	anytimeGuaranteeSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR280	finiteHorizonGuaranteeSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR281	finiteHorizonGuaranteeSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR282	finiteHorizonGuaranteeSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR283	statisticalGuaranteeSemanticsSufficientSets_ eq_of_outputSemanticsEquivalent Paper4dFrontier/Realizability.lean
FR284	statisticalGuaranteeSemanticsRelevantSet_eq_ of_outputSemanticsEquivalent Paper4dFrontier/Realizability.lean	FR285	statisticalGuarantee_classifier_agrees_on_ closureEquivalent_of_correctOnDomain_of_ transfer Paper4dFrontier/StatisticalSemanticsApplications.lean
FR286	no_correctOnDomain_statisticalGuarantee_ classifier_of_orbit_gap_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean	FR287	statisticalGuarantee_correct_classifier_ onDomain_iff_no_orbitGapOn_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean
FR288	opt_eq_of_uniformApprox_of_ uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean	FR289	sufficientSets_eq_of_uniformApprox_of_ uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean
FR290	relevantSet_eq_of_uniformApprox_of_ uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean	FR291	randomizedGuaranteeSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR292	randomizedGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR293	randomizedGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR294	randomizedGuarantee_classifier_agrees_on_closureEquivalent_of_correctOnDomain_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean	FR295	no_correctOnDomain_randomizedGuarantee_classifier_of_orbit_gap_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean
FR296	randomizedGuarantee_correct_classifier_onDomain_iff_no_orbitGapOn_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean	FR297	transferredSemantics_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/StatisticalSemanticsApplications.lean
FR298	no_correctOnDomain_transferredSemantics_classifier_of_orbit_gap Paper4dFrontier/StatisticalSemanticsApplications.lean	FR299	transferredSemantics_correct_classifier_onDomain_iff_no_orbitGapOn Paper4dFrontier/StatisticalSemanticsApplications.lean
FR300	decisionEquiv_iff_of_uniformApprox_of_uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean	FR301	isMinimalSufficient_iff_of_uniformApprox_of_uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean
FR302	outputSemanticsExactRelevanceProfile_eq_of_outputSemanticsEquivalent Paper4dFrontier/Realizability.lean	FR303	outputSemanticsExactRelevanceProfile_eq_totalizedLawDecisionProblem Paper4dFrontier/Realizability.lean
FR304	agreeOn_univ_iff_eq_of_finiteIndicatorCoordinateSpace Paper4dFrontier/Realizability.lean	FR305	finite_outputSemantics_allCoordinatesSufficient Paper4dFrontier/Realizability.lean
FR306	finite_outputSemantics_realized_by_exactCertification Paper4dFrontier/Realizability.lean	FR307	agreeOn_univ_iff_eq_of_singletonIdentityCoordinateSpace Paper4dFrontier/Realizability.lean
FR308	outputSemantics_admits_coordinatePresentation Paper4dFrontier/Realizability.lean	FR311	encodable_stateSpace_admits_countableBooleanPresentation Paper4dFrontier/Realizability.lean
FR312	encodable_discreteMeasurable_stateSpace_admits_measurable_countableBooleanPresentation Paper4dFrontier/Realizability.lean	FR313	encodable_discreteTopological_stateSpace_admits_continuous_countableBooleanPresentation Paper4dFrontier/Realizability.lean
FR314	agreeOn_univ_iff_eq_of_finiteBinaryCoordinateSpace Paper4dFrontier/Realizability.lean	FR315	finite_exactSpecification_admits_lowDimBooleanPresentation Paper4dFrontier/Realizability.lean
FR316	IdentityOutputClosureSpec.classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR317	IdentityOutputClosureSpec.no_correctOnDomain_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR318	hypothesisOutput_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR319	estimatorOutput_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR320	policyOutput_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR321	randomizedProcedure_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR322	no_correctOnDomain_hypothesisOutput_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR323	no_correctOnDomain_estimatorOutput_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR324	no_correctOnDomain_policyOutput_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR325	no_correctOnDomain_randomizedProcedure_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR326	TransportedOutputClosureSpec.classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR327	TransportedOutputClosureSpec.no_correctOnDomain_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR328	IdentityOutputClosureSpec.correctOnDomain_iff_correctOnDomain_transport Paper4dFrontier/ComputeCostExternalOutputs.lean	FR329	representationRelativeHypothesis_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR330	representationRelativeEstimator_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR331	representationRelativePolicy_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR332	representationRelativeRandomizedProcedure_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR333	no_correctOnDomain_representationRelativeHypothesis_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
FR334	no_correctOnDomain_representationRelativeEstimator_ classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR335	no_correctOnDomain_representationRelativePolicy_ classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR336	no_correctOnDomain_representationRelativeRandomizedProcedure_ classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR340	booleanPayloadTransfer Paper4dFrontier/Realizability.lean
FR341	predicateTransfer Paper4dFrontier/Realizability.lean	FR342	sufficiency_is_relation_refinement Paper4dFrontier/Realizability.lean
FR343	relevance_is_erased_failure_of_refinement Paper4dFrontier/Realizability.lean	FR345	exactSemanticsQuotient_universal_ characterization Paper4dFrontier/Realizability.lean
FR346	exactnessMeansExactAgreementWithValidity Paper4dFrontier/Realizability.lean	FR347	closure_sound_package_no_go_uses_only_ closureLawInvariant Paper4dFrontier/FullBinaryPairwiseDomain.lean
FR348	fullBinaryPairwiseDomain_closure_closed Paper4dFrontier/FullBinaryPairwiseDomain.lean	FR349	fullBinaryPairwiseDomain_contains_four_ obstruction_families Paper4dFrontier/FullBinaryPairwiseDomain.lean
FR350	no_correct_tractability_classifier_on_ fullBinaryPairwiseDomain Paper4dFrontier/FullBinaryPairwiseDomain.lean	FR351	ExactCorrectnessSpecification.universal_scope_ over_rigorously_specified_problems Paper4dFrontier/Realizability.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 2.45: Exact Approximation Semantics Transfer	FR234, FR233, FR232
Corollary 2.26: Boolean Payload Transfer	FR340
Corollary 2.6: Constant-Optimizer Collapse Forces Trivial Certification	FR52, FR53
Corollary 6.18: Domain Restriction Helps Only by Removing Orbit Gaps	FR252, FR254, FR255, FR253, FR256
Corollary 2.51: Every State Equivalence Relation Is Realizable as Exact Output Semantics	FR240, FR237
Corollary 2.20: Compute-Cost Version: External Output Objects	FR316, FR328, FR317, FR326, FR327, FR319, FR318, FR323, FR322, FR324, FR325, FR320, FR321
Corollary 3.3: Finite Basis for the Current Positive Landscape	FR3, FR29
Corollary 6.15: No Exact Classifier on the Full Binary Pairwise Domain	FR350
Corollary 6.13: No Correct Tractability Classifier	FR197, FR199, FR193, FR198
Corollary 6.17: Orbit-Gap Completeness on Closure-Closed Domains	FR210, FR212, FR213, FR211
Corollary 2.19: Compute-Cost Version: Canonical Payload and Search Tasks	FR260, FR259, FR262, FR261
Corollary 2.27: Predicate Transfer	FR341
Corollary 2.43: Randomized-Output Guarantee Semantics Inherit the Closure-Orbit Consequences	FR295, FR293, FR292, FR291, FR294, FR296
Corollary 2.42: Randomized-Output Guarantee Semantics Transfer	FR293, FR292, FR291
Corollary 5.4: Realizability Alone Cannot Be the Frontier	FR7, FR8
Corollary 6.12: Finite Structural Admissibility Packages	FR194
Corollary 2.30: Arbitrary Exact Output Semantics Transfer	FR231, FR230, FR229
Corollary 2.3: Relevance Is Failure of Erased Sufficiency	FR49, FR50

Paper claim	**Lean handle**
Corollary 2.33: Relevance Is Erased Failure of Refinement	FR343
Corollary 2.28: Nonempty Set-Valued Payload Transfer	FR225, FR224, FR223
Corollary 2.47: Arbitrarily Small Uniform Perturbations Can Flip Relevance	FR264
Corollary 2.48: Arbitrarily Small Uniform Perturbations Can Flip Sufficiency	FR267
Corollary 2.44: Statistical Guarantee Semantics Inherit the Closure-Orbit Consequences	FR279, FR278, FR277, FR282, FR281, FR280, FR295, FR286, FR298, FR288, FR270, FR269, FR268, FR293, FR292, FR291, FR294, FR296, FR273, FR272, FR271, FR290, FR284, FR283, FR285, FR287, FR276, FR275, FR274, FR289, FR297, FR299
Corollary 2.41: Statistical Guarantee Semantics Transfer	FR279, FR278, FR277, FR282, FR281, FR280, FR270, FR269, FR268, FR273, FR272, FR271, FR276, FR275, FR274
Corollary 2.4: Certification Statistics Factor Through the Quotient Relation	FR41, FR43, FR44, FR42
Corollary 2.2: Sufficiency Is Relation Refinement	FR48
Corollary 2.29: Arbitrary Set-Valued Payload Transfer via Failure Tokens	FR228, FR227, FR226
Corollary 2.21: Compute-Cost Version: Representation-Relative Output Objects	FR328, FR326, FR327, FR334, FR333, FR335, FR336, FR330, FR329, FR331, FR332
Corollary 2.35: Universal Scope Over Rigorously Specified Problems	FR351
Proposition 2.22: The Finite Structural Class Is Nonempty	FR200
Proposition 2.8: Positive Affine Utility Reparameterizations Are Invisible	FR13, FR16, FR15, FR14
Proposition 2.46: Approximate Relevance and Sufficiency Claims Need Explicit Stability Control	FR266, FR263
Proposition 6.1: Binary Pairwise Symmetry Dichotomy	FR120, FR118, FR119
Proposition 2.24: Bounded-Pattern Predicates Stabilize Above a Finite Action Bound	FR215, FR216, FR214
Proposition 2.36: Canonical Exact Relevance Profile	FR302, FR303
Proposition 2.15: Closure Operations Preserve Exact Certification	FR57, FR17, FR18, FR60, FR56, FR59, FR84, FR16, FR55, FR58, FR83, FR15, FR19, FR20, FR85
Proposition 2.39: Countable Exact Specifications Admit Countable Boolean Presentations	FR312, FR313, FR311
Proposition 6.2: Decision-Relevant Binary Pairwise Dichotomy	FR121, FR123, FR124, FR122
Proposition 4.2: Degenerate Positive Cases	FR1, FR6
Proposition 2.1: Exact Certification Depends Only on the Decision Quotient Relation	FR32, FR33, FR31, FR30, FR37, FR51, FR35, FR34
Proposition 2.23: Bounded Distinct Action Profiles Compress to Bounded Actions	FR206, FR207, FR205, FR209, FR208
Proposition 2.9: Duplicate Actions and Duplicate States Preserve Certification	FR71, FR68, FR54, FR70, FR56, FR69, FR55, FR73, FR72
Proposition 2.7: Explicit Enumeration Is Parameter-Dependent, Not Structural	FR66, FR67
Proposition 2.37: Every Exact Specification Admits a Coordinate Presentation	FR307, FR308
Proposition 2.31: Exactness Means Exact Agreement With Validity	FR346

Paper claim	Lean handle
Proposition 3.4: Explicit Primitive Basis	FR29
Proposition 3.2: Explicit Inventory	FR26, FR25, FR3, FR28, FR27
Proposition 2.38: Finite Exact Specifications Admit Coordinate Presentations	FR304, FR305, FR306
Proposition 2.40: Finite Exact Specifications Admit Low-Dimensional Boolean Presentations	FR314, FR315
Proposition 3.5: All Independent Core Mechanisms Are Nondegenerate	FR108, FR106, FR112, FR101, FR105, FR113
Proposition 6.14: Full Binary Pairwise Domain Is Closure-Closed	FR348, FR349
Proposition 2.49: Global Approximation Stability Under Uniform Strict Gaps	FR300, FR301, FR295, FR298, FR288, FR293, FR292, FR291, FR294, FR296, FR290, FR289, FR297, FR299
Proposition 3.9: Current Positive Interfaces Factor Through Role Classification	FR4
Proposition 2.12: Invariance Alone Is Too Weak	FR22, FR21, FR23
Proposition 3.6: Weak Tree Ordering Does Not Imply Width One	FR115, FR114
Proposition 4.1: Lifted Cases Introduce No New Primitive	FR5
Proposition 6.11: Logical Dependency of the No-Go	FR347
Proposition 2.14: Unrestricted Predicates Trivialize Exact Characterization	FR145
Proposition 6.16: Orbit-Gap Completeness for Exact Classification	FR203, FR201, FR202, FR204
Proposition 6.3: Orbit-Gap Template	FR185
Proposition 3.8: Parent-Tree Structure Gives Width-One Decompositions	FR110, FR100, FR111
Proposition 2.25: Deterministic Payload Transfer	FR219, FR220, FR222, FR218, FR221, FR217
Proposition 2.11: Quotient Size Is Unbounded Under Realizability	FR24, FR38
Proposition 5.3: The Realized Quotient Is Exactly the Label Range	FR39, FR40
Proposition 2.10: Relabeling Invariance Is Forced by Exact Certification	FR17, FR18, FR19, FR20
Proposition 2.50: Exact Semantic Claims Depend Only on Admissible-Output Equivalence	FR236, FR235
Proposition 2.52: Semantically Extensional Claims Factor Through the Exact-Semantics Quotient	FR242, FR241, FR244, FR243
Proposition 5.2: Every Equivalence Relation Is Optimizer-Realizable	FR47, FR46, FR45
Proposition 2.13: Independent Summary Frameworks Converge on the Same Structural Core	FR178, FR181, FR182, FR180, FR179
Proposition 2.32: Sufficiency Is Relation Refinement	FR342
Proposition 3.7: Cyclic Dependencies Recover Hardness	FR110, FR100, FR111, FR115
Proposition 2.5: Zero-Distortion Summaries Refine the Optimizer Quotient	FR177, FR176
Theorem 6.9: Finite-Structural Collapse Across the Four Obstruction Families	FR190
Theorem 6.10: Closure-Sound Package No-Go	FR193
Definition 2.17: Finite Structural Predicate	FR197, FR199, FR198, FR249, FR248, FR245, FR246, FR247
Theorem 3.1: Explicit Partition of the Current Positive Landscape	FR26, FR25, FR1, FR2, FR28, FR27

Paper claim	Lean handle
Theorem 6.5: Dominant-Pair Finite-Structural No-Go	FR186
Theorem 2.34: Exact Semantics Quotient Universality	FR345
Theorem 6.7: Ghost-Action Finite-Structural No-Go	FR188
Theorem 5.1: Every Labeling Kernel Is Optimizer-Realizable	FR7, FR8
Theorem 6.6: Margin-Masking Finite-Structural No-Go	FR187
Theorem 6.8: Offset Finite-Structural No-Go	FR189
Theorem 2.18: Compute-Cost Version: Optimizer Computation	FR248

Auto summary: mapped 83/83 (full=83, derived=0, unmapped=0).